

Introduction to Machine Learning

The objective of this assignment is to build an end-to-end machine learning pipeline using Python and scikit-learn. You will train a Logistic Regression model to predict whether a patient has ever experienced any type of food allergy based on their demographic information and medical history.

Dataset

Load the data directly into your notebook using the following URL:

https://claproduction.s3.eu-west-3.amazonaws.com/DataScience/3_classical_ml/food-allergy-analysis-Zenodo.csv

Instructions

Step 1: Data Preparation and Feature Engineering

- Load the CSV file into a pandas dataframe.
- **Target Variable Creation:** The dataset records specific allergies across multiple columns (e.g., peanut, milk, shellfish). You must engineer a single binary target column named *has_food_allergy*. Set this to 1 if a patient has tested positive in any of the specific food allergy columns, and 0 otherwise.
- **Feature Selection:** Define your predictor variables. Exclude all direct food allergy columns from your features to prevent data leakage. Use demographic data (like age, gender, race) and related medical conditions (like asthma, atopic dermatitis, and allergic rhinitis).

Step 2: Preprocessing Pipeline

- Split the dataset into a training set and a testing set using an 80:20 split ratio.
- Create a preprocessing pipeline using *ColumnTransformer*.
- **Numerical Features:** Handle missing values using a median imputer, then scale the data using *StandardScaler*.
- **Categorical Features:** Handle missing values with a constant imputer (e.g., fill with 'missing'), then encode the text data into numbers using *OneHotEncoder*.

Step 3: Model Training and Evaluation

- Initialize a *LogisticRegression* classifier. *Hint: Because allergy cases make up a smaller percentage of the dataset, consider using the `class_weight='balanced'` parameter to help the model pay attention to the minority class.*

- Combine your preprocessor and classifier into a single scikit-learn *Pipeline* and fit it to your training data.
- Predict the outcomes for your testing data.
- Print the confusion matrix and the classification report.

Submission Guidelines

- Submit your completed Python code as a Jupyter Notebook (.ipynb) or Python script (.py file).
- Ensure your code is well-commented, organized, and runs without errors.
- At the bottom of your file, include a short text block or comment summarizing your model's performance. Briefly explain whether you think precision or recall is more important when predicting a medical condition like a food allergy.